Ruben Valenzuela

Daedra Minigan

Benjamin Lacount II

Professor Viswanatha Rao

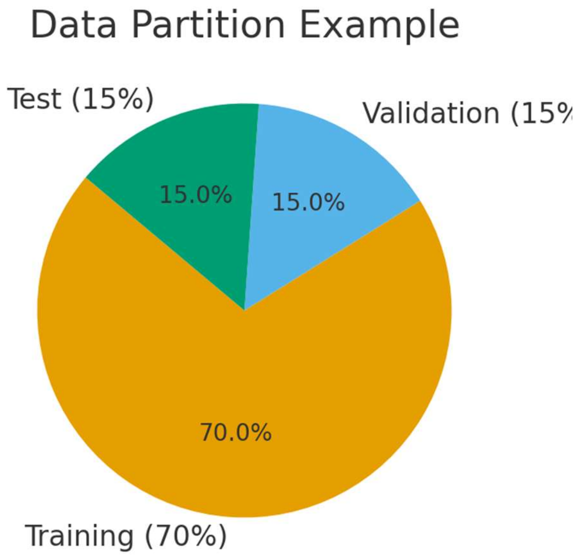
ITAI-2376-Deep Learning Artificial Intel

09 September 2025

Machine Learning Concepts Review

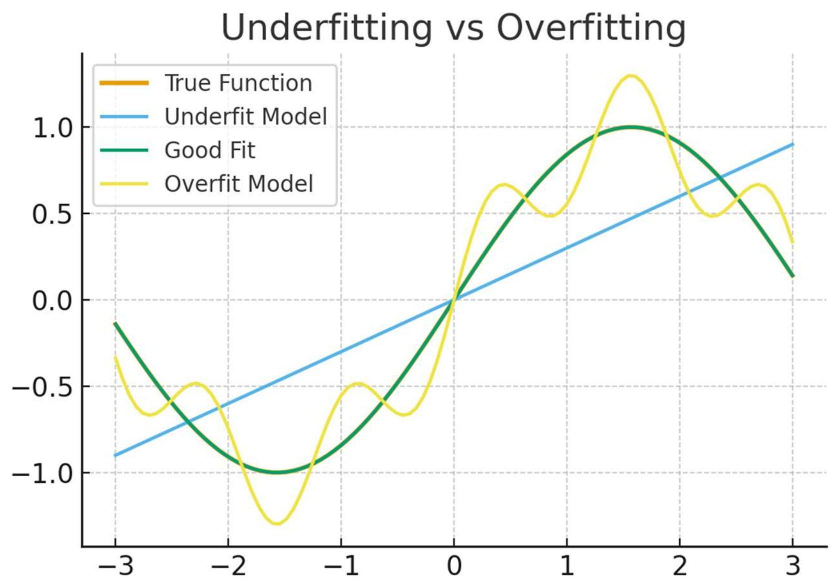
1. Training vs Validation Phase

In machine learning, the training phase involves fitting the model to data by minimizing a loss function. The validation phase evaluates the model on unseen data to measure generalization. Typically, datasets are partitioned into training, validation, and testing sets (e.g., 70%/15%/15%). Partitioning prevents information leakage and ensures that model performance reflects real-world outcomes. If no partition is made, the model may memorize the training data and appear accurate but fail in practice.



2. Underfitting vs Overfitting

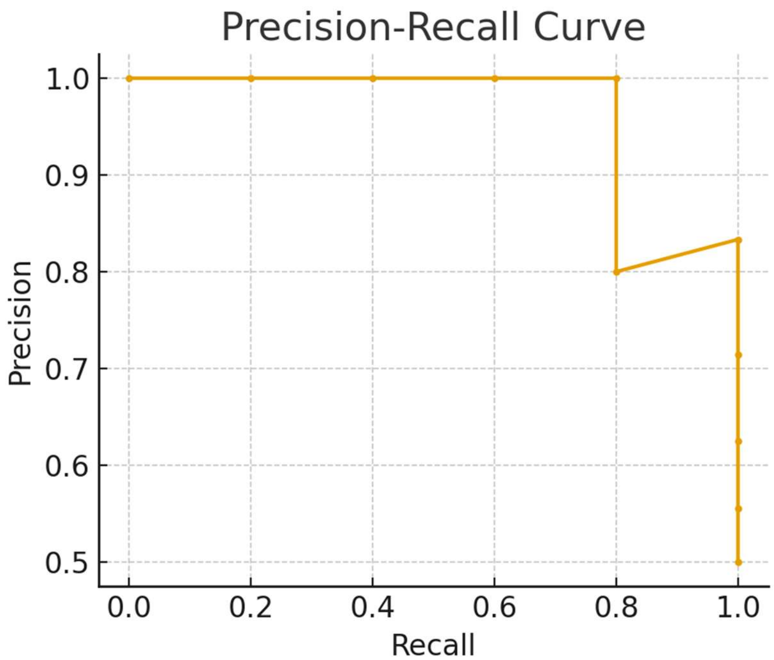
Underfitting occurs when the model is too simple to capture the data’s patterns, while overfitting happens when the model memorizes noise instead of generalizing. Real-life example: In predicting housing prices, an underfit model may only consider square footage, ignoring other factors, while an overfit model memorizes exact house IDs. Another example: In medical diagnosis, underfitting may miss key symptoms, while overfitting may rely on irrelevant details, producing unstable predictions.



Minimizing underfitting can be done by increasing model complexity or training longer. To minimize overfitting, one can use regularization, dropout, or more data.

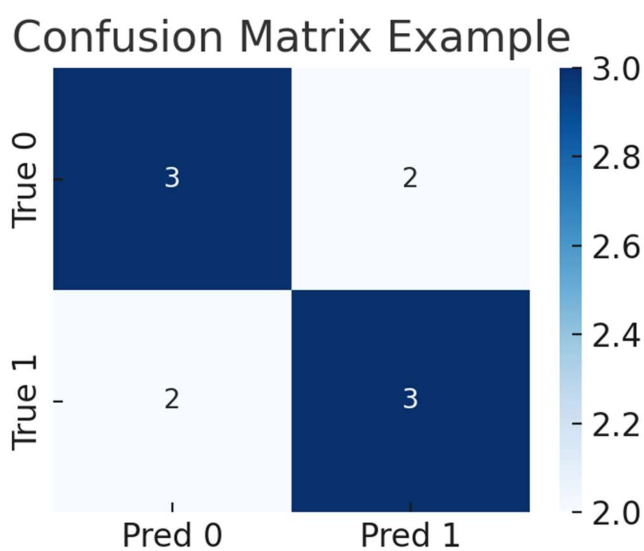
3. Classification vs Regression

Classification involves predicting discrete labels. Examples: determining if an email is spam (yes/no), or classifying handwritten digits. Regression involves predicting continuous values. Examples: predicting house prices or stock market returns. Validation parameters differ: accuracy measures overall correctness, while recall focuses on identifying positive cases. Accuracy is intuitive but may be misleading in imbalanced datasets.



4. Confusion Matrix

A confusion matrix is a 2x2 table summarizing classification results: True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN). It provides deeper insights than accuracy alone, showing where errors occur.



For example, if a medical test has high false negatives, it may fail to detect sick patients, making recall a critical metric. Confusion matrices allow practitioners to evaluate such trade-offs.

Conclusion

In summary, training and validation phases ensure models generalize to unseen data, while underfitting and overfitting represent extremes of model performance. Classification and regression are core ML tasks with distinct evaluation metrics, and confusion matrices provide detailed diagnostic insights. Together, these concepts form the foundation of applied machine learning.